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Assignment 2

Question 1

$26 = 16 + 8 + 2 = 2^4 + 2^3 + 2^1$

```
mov bx, ax
shl bx, 4
mov cx, bx

mov bx, ax
shl bx, 3
add cx, bx

shl ax, 1
add cx, ax
```

Question 2

```
mov eax, val
rol eax, 8
rol eax, 8
rol eax, 8
rol eax, 8

ror eax, 32
```

Question 3

```
.data
    val dword 01234SABCDh
.code
    MAIN PROC
        mov eax, [val]
        rol eax, 16
        rol eax, 16
        mov eax, [val]
        ror eax, 16
        ror eax, 16
```

Question 9

```
.data
A Byte "A",0
B Byte "B",0
C Byte "C",0
F Byte 'F',0
```

```
.code
MAIN PROC
 call readint
cmp eax, 80
jae is_A
cmp eax, 60
jae is_B
cmp eax, 40
jae is_C
```

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Question 4

a) mov ax, timestamp
and al, ~~HH000000 h~~ 00011111h
mov bseconds, al

b) mov ax, timestamp
shr ax, 5
and al, 00111111h
mov bminutes, al

Question 5

Dx = 1A63h Ax = 10A0h CF = 1

Question 6

a. test ax, ax
b. test ax, 1
c. test Dx, Dx
d. test Dx, 8000h
e. test BL, Bl

Question 7

10h → 0001 0000 → 16
5 + 16 + 1
Dx = 22

Question 8

1. mov offset int edi/esi
   mov esi, offset val1
   mov edi, offset val2
2. inc esi & edi
3. store result in result not in val1

Question 9 cont....

```
jmp is_A

is_A: mov al, 'A'
      jmp display
is_B: mov al, 'B'
      jmp display
is_C: mov al, 'C'
      jmp display
_F:   mov al, 'F'
display: call writechar
exit
main ENDP
END Main
```

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Question 10

EAX = 3 EBX = 10

Question 11

```
mov eax, num1
cmp eax, num2
jle elseif
mov eax, num3
cmp eax, num4
je else if
mov result, num1
add result, num3
                jmp end-if
else_if:
  cmp num2, num4
  jne else
  mov result, num2
  sub result, num1
                jmp end-if
else:
  result, 0
end-if:
```

Question 12

```
while:
  mov eax, op1
  cmp eax, op2
  jg end_while
  cmp eax, x
  jl else
  cmp eax, y
  jg else
  add z, 10
  jmp end_if
else:
  sub z, 10
end_if:
  dec op1
loop while
end_while:
```

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